

Worksheet

Do the following series converge or diverge? Be sure to state any tests used and show all work needed apply them.

1. $\sum_{k=0}^{\infty} \frac{4}{5k-3}$

2. $\sum_{k=1}^{\infty} \frac{1}{k^2+5}$

3. $\sum_{k=0}^{\infty} \frac{4}{k!}$

4. $\sum_{k=1}^{\infty} \frac{3^{2k+1}}{2^k 7^{k-1}}$

5. $\sum_{k=1}^{\infty} \frac{3^k}{2^k + 5k^2}$

6. $\sum_{k=1}^{\infty} \frac{\sqrt{k}+4}{k^2-2}$

7. $\sum_{k=2}^{\infty} \left(1 + \frac{2}{k}\right)^k$

8. $\sum_{k=1}^{\infty} \frac{3^k}{(2k+1)!}$

9. $\sum_{k=1}^{\infty} \frac{3^k}{(2k)^k}$

10. $\sum_{k=2}^{\infty} \frac{3 \ln(k+1)}{\sqrt{k}}$

11. $\sum_{k=1}^{\infty} \frac{3^k}{(2k+1)!}$

12. $\sum_{k=1}^{\infty} \frac{\sqrt{k}}{3^k}$

13. Find all values of x so that $\sum_{k=2}^{\infty} \frac{|x-3|^k}{4^k}$ converges.

14. Find all values of x so that $\sum_{k=2}^{\infty} \frac{|x|^k}{3^k}$ converges.

15. State and explain the Zero-Infinity Comparison Test.